Engineering Research
Express

INCLUSIVE PUBLISHING
TRUSTED SCIENCE

PURPOSE-LED
PUBLISHING™

PAPER

A multi-scale self-attention method with uncertainty quantification for probabilistic remaining useful life prediction of rotating machinery

To cite this article: Jiashuo Du and Jinze Fu 2025 *Eng. Res. Express* **7** 035543

View the article online for updates and enhancements.

You may also like

- A variational autoencoder based lightweight physics-informed neural network for remaining useful life prediction
Yonghao Miao, Yu Xia and Jiantao Chang
- Remaining useful life prediction with uncertainty quantification based on multi-scale attention-based physics-informed neural network
Baimao Lei, Yu Wang, Mingjian Li et al.
- Remaining useful life prediction based on time-series features and conformalized quantile regression
Song Mao, Xiaofeng Li and Boyang Zhao

This content was downloaded from IP address 218.197.145.100 on 04/11/2026 at 12:46

Engineering Research Express

PAPER

A multi-scale self-attention method with uncertainty quantification for probabilistic remaining useful life prediction of rotating machinery

RECEIVED
5 March 2025

REVISED
3 July 2025

ACCEPTED FOR PUBLICATION
11 July 2025

PUBLISHED
7 August 2025

Jiashuo Du[1,*] **and Jinze Fu**[2]

[1] School of Mechanical Engineering, Inner Mongolia University, Hohhot 010021, People's Republic of China
[2] School of Mathematical Sciences, Inner Mongolia University, Hohhot 010021, People's Republic of China
[*] Author to whom any correspondence should be addressed.

E-mail: 0223123723@mail.imu.edu.cn (**J Du**) **and** 3187265823@qq.com (**J Fu**)

Keywords: multi-scale learning, self-attention, RUL prediction, Monte Carlo dropout, uncertainty quantification

Abstract

The prediction of remaining useful life (RUL) for rotating machinery plays a critical role in Prognostic and Health Management (PHM) of industrial equipment. However, existing machine learning-based RUL prediction methods often neglect multi-scale feature learning and fail to quantify prediction uncertainty, which limits their practical application in real-world scenarios. To address these limitations, this paper proposes a Multi-Scale Self-Attention method with Uncertainty Quantification (MSAUQ). First, a multi-scale attention module is designed and integrated into the encoder of MSAUQ to effectively extract hierarchical features from time-series data while capturing global dependencies. Subsequently, by incorporating an approximate Bayesian framework and kernel density estimation (KDE), the model enables probabilistic density estimation of RUL predictions, thereby achieving uncertainty quantification. Experimental results demonstrate that the proposed MSAUQ exhibits superior performance across all four subsets of the NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset and effectively quantifies prediction uncertainty, providing both accurate RUL estimations and interpretable confidence intervals for maintenance decision-making.

Abbreviations

The following abbreviations are used in this manuscript

UQ	Uncertainty Quantification
PDF	Probability Density Function
PHM	Prognostics and Health Management
RUL	Remaining Useful Life
C-MAPSS	Commercial Modular Aero-Propulsion System Simulation
KDE	Kernel Density Estimation
CIs	Confidence Intervals
MSSA	Multi-Scale Self-Attention
MSAUQ	Multi-Scale Self-Attention Method with Uncertainty Quantification
MC	Monte Carlo
PE	positional encoding

© 2025 IOP Publishing Ltd. All rights, including for text and data mining, AI training, and similar technologies, are reserved.

No PE	No Positional Encoding
Learnable PE	Learnable Absolute Positional Encoding
Embedding PE	Embedding-based Learnable Positional Encoding
Sinusoidal PE	Sinusoidal Positional Encoding
BNNs	Bayesian Neural Networks

1. Introduction

The prediction of remaining useful life (RUL) is a core task in Prognostics and Health Management (PHM) [1], aiming to assess the remaining operational lifespan of equipment through real-time monitoring of its operating conditions. This enables the early identification of potential failures and optimization of maintenance decisions [2, 3]. This task is not only critical for enhancing equipment reliability and reducing maintenance costs but also plays a vital role in ensuring production safety and extending equipment life. Therefore, efficiently and accurately achieving RUL prediction has become a key challenge in the fields of intelligent manufacturing and equipment maintenance [4, 5].

Currently, RUL prediction methods can be categorized into three types: physics-based methods, statistical methods, and deep learning methods [2]. Physics-based methods construct physical models of equipment to simulate the degradation process under varying operating conditions and predict RUL based on these models [6, 7], While these methods provide clear explanations of failure mechanisms [8], their applicability is limited by the complexity of equipment and diverse environmental conditions. Additionally, building accurate models for complex systems often incurs high computational costs [9]. Moreover, the accuracy of physics-based methods is highly sensitive to model assumptions and parameter selection [10], and maintaining stability in dynamic environments remains a challenge [11].

Statistical methods, on the other hand, rely on stochastic process theory and probabilistic modeling of the degradation process to estimate RUL [12]. These methods assume the degradation process follows a specific statistical distribution or stochastic model, deriving the RUL and its probability density function (PDF). For example, Fan *et al* [13] used SPC to adaptively evaluate bearing degradation states, constructing different Wiener process-based degradation models to describe the degradation path and quantify the uncertainty of RUL predictions. Lv *et al* [14] employed dynamic entropy-weighted minimum quantization error (DEWMQE) as the initial health indicator (HI) and used linear multi-fractional Lévy stable motion (LMLSM) to establish degradation models. Wang *et al* [15] proposed a degradation angle as the HI and constructed a two-stage Wiene r process-based degradation model for RUL prediction. Although statistical methods are more flexible compared to physics-based methods [16], they remain dependent on external assumptions regarding HI selection and degradation path modeling [17]. When dealing with highly nonlinear or complex degradation patterns, statistical methods often fail to provide sufficiently accurate predictions [12].

In recent years, deep learning methods have emerged as a dominant approach for RUL prediction, owing to their strong feature extraction and efficient model learning capabilities [18]. Popular deep learning models, such as those based on CNN [19], RNN [3], and LSTM [20], can effectively handle nonlinear degradation processes by automatically extracting complex features from large amounts of historical data [21]. For instance, Zhang *et al* [22] integrated particle filtering, temporal attention mechanisms, and BiGRU to incorporate battery degradation mechanisms into RUL prediction. Shi *et al* [23] proposed a lightweight model combining dual attention and LSTM to address deficiencies in traditional attention mechanisms for sequential degradation feature recognition. Zhang *et al* [24] leveraged temporal convolution and attention mechanisms for bearing RUL prediction, achieving promising results.

Although progress has been made in RUL prediction using deep learning [25], most existing methods focus on feature learning at a single scale, neglecting the multi-level and multi-scale features inherent in equipment degradation processes [26]. In practice, the health status of equipment often exhibits cross-scale characteristics, where degradation processes at different scales may reveal distinct failure patterns [27, 28]. Feature learning at a single scale cannot fully capture these subtle changes [29], leading to limitations in prediction accuracy and robustness when handling complex and heterogeneous degradation modes. Therefore, incorporating multi-scale learning methods [30] to integrate features from different levels and uncover the variations in equipment health status across multiple scales can enable models to comprehensively capture degradation processes, thereby improving prediction accuracy and robustness [31].

Additionally, most existing RUL prediction methods focus on point estimates of RUL [32], overlooking the quantification of prediction uncertainty [33, 34]. In reality, the health status of equipment is influenced by multiple factors, such as sensor noise, environmental variations, and unknown failure modes, which introduce uncertainty into RUL predictions. Traditional point estimation methods cannot fully reflect these uncertainties

[32], potentially compromising the reliability of maintenance decisions [35]. In practical engineering applications, considering the confidence interval or probability distribution of prediction results is essential [36], as it provides more meaningful risk assessments in complex and high-risk industrial scenarios [37].

Based on the discussion above, current RUL prediction methods face two main challenges: (1) the lack of multi-scale feature learning, as existing methods often rely on single-scale or fixed-window feature extraction, limiting their ability to extract and process multi-level and multi-scale features. This constraint hinders the precise prediction of equipment behavior throughout its lifecycle; (2) the neglect of uncertainty quantification (UQ), as most methods focus on point estimates while failing to provide uncertainty analysis of prediction results. This limitation not only reduces the trustworthiness of models in predictive maintenance scenarios but also impairs the reliability and accuracy of maintenance decisions.

To address these issues, we propose a Multi-Scale Self-Attention method with Uncertainty Quantification (MSAUQ) which combines a multi-scale self-attention mechanism with UQ. By integrating multi-scale learning with self-attention mechanisms, MSAUQ extracts multi-level features from time-series data while capturing global dependencies. Additionally, an approximate Bayesian framework is introduced, combined with kernel density estimation (KDE), to derive the probability density distribution of prediction results, enabling UQ. Experimental results demonstrate that the proposed method achieves excellent prediction performance on the C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset. The main contributions of this paper are summarized as follows:

1. A multi-scale self-attention module (MSSA) is proposed, combining multi-scale learning and self-attention mechanisms to extract local features through multi-scale learning while leveraging self-attention to capture global dependencies, thereby enhancing feature representation capabilities for RUL prediction tasks.
2. Based on the approximate Bayesian framework and KDE, probabilistic RUL prediction is achieved. MSAUQ provides both point estimates and UQ results, enabling the estimation of PDF and 95% confidence intervals (CIs) for prediction results.
3. Extensive experiments are conducted on the C-MAPSS dataset to validate MSAUQ's performance. The results demonstrate that MSAUQ achieves ideal RMSE and SCORE performance across all four subsets of the C-MAPSS dataset.

The remainder of this paper is organized as follows: section 2 provides the relevant background for the methods used in this study. Section 3 presents the detailed framework of the proposed model and its components. Section 4 describes the experimental setup and results on the C-MAPSS dataset, including analyses of prediction errors and uncertainty. Section 5 concludes the paper and outlines future research directions.

2. Background

2.1. Multi-scale learning

In the RUL prediction task, multi-scale learning refers to jointly learning the features of equipment health states across multiple characteristic scales to capture degradation information at different levels and temporal scales. Since the degradation patterns of equipment during actual operation often exhibit cross-scale characteristics—where degradation processes at different time scales may reveal distinct fault modes and health state evolution patterns—a single-scale learning approach is insufficient to capture these multi-level insights comprehensively. By adopting a multi-scale learning method, features from different scales can be integrated, thereby improving the model's ability to predict complex degradation processes.

In RUL prediction tasks, multi-scale learning is typically implemented through multi-level feature extraction modules. These modules analyze operational data of equipment across various scales, capturing health state changes from short-term to long-term perspectives. Assuming the health indicator sequence of the equipment is $X = \{x_1, x_2,\ldots,x_T\}$, where $x_t \in R^n$ represents sensor data or health indicators obtained at time step t, n is the data dimensionality, and T is the sequence length. Traditional single-scale learning methods often rely on global modeling of such time-series data but neglect the degradation features at different time scales.

To enable learning across different scales, a scale transformation function $\mathscr{S}_k$ can be introduced, which converts the original sequence X into representations at different scales $X^{(k)}$. Formally, $X^{(k)}$ can be defined as:

$$X^{(k)} = \mathscr{S}_k(X), \quad k \in \{1, 2,\ldots,K\} \tag{1}$$

Here, k represents the number of scales considered, and $\mathscr{S}_k$ is a scale transformation operation, which may include methods such as sliding windows, downsampling, or feature extraction across different time scales using convolutional kernels. Through these scale transformation operations, the health information of the equipment

is represented as multiple feature sequences $X^{(1)}$, $X^{(2)}$,…,$X^{(K)}$, each revealing the degradation trends of the equipment at a different time scale.

On this basis, the final health state representation h can be achieved by weighted aggregation of multi-scale feature representations:

$$h = \sum_{k=1}^{K} w_k \cdot X^{(k)} \tag{2}$$

where w_k is the weight coefficient, which can be learned during training. This process ensures that information from different scales is effectively combined to comprehensively reflect the degradation trends of the equipment. In this way, the feature information from different scales complements one another, helping the model overcome the limitations of single-scale approaches and achieve better RUL prediction performance.

2.2. Uncertainty quantification

In the RUL prediction task, the goal of UQ is to measure and characterize the inherent uncertainty in the model's predictions. The uncertainty of a model mainly arises from factors such as incomplete data, simplified model assumptions, system complexity, and external noise. Suppose we have a set of historical monitoring data $D = \{X, Y\}$, where X is the feature matrix of the equipment, and Y is the corresponding RUL labels. Based on this data, we aim to predict the equipment's RUL at a future time t. In deep learning models, the health state of equipment can be represented by a parameterized model $f(x; \theta)$, where θ represents the model parameters and x is the input features. The model's prediction $\hat{y}$ depends on these parameters, and due to data incompleteness and noise, the model output $\hat{y}$ contains uncertainty. Therefore, we treat the model's prediction $\hat{y}$ as a random variable, whose distribution can be expressed as the conditional probability $p(\hat{y}|x, D)$. This indicates the need to infer the probability distribution of RUL given the input data and historical information.

Under the Bayesian framework, UQ involves inferring the posterior distribution of the model parameters θ, which is given by:

$$p(\theta|\, D) = \frac{p(D|\, \theta)p(\theta)}{p(D)} \tag{3}$$

where $p(D|\, \theta)$ is the likelihood function, representing the probability of observing the data given the model parameters; $p(\theta)$is the prior distribution, reflecting prior knowledge about the parameters θ; $p(D)$ is the evidence, representing the overall probability of the data.

However, the posterior distribution $p(\theta|\, D)$ is often intractable to solve analytically, especially in high-dimensional neural networks. Therefore, we can use variational inference as an approximation method. Variational inference introduces a variational distribution $q(\theta)$ to approximate the true posterior distribution $p(\theta|\, D)$. The goal of variational inference is to find the optimal variational distribution by minimizing the Kullback-Leibler (KL) divergence between $q(\theta)$ and the true posterior $p(\theta|\, D)$. This objective can be expressed as:

$$\mathcal{L}(q) = KL(q(\theta)\|p(\theta|\, D)) = E_{q(\theta)}\left[\log \frac{q(\theta)}{p(D|\, \theta)p(\theta)}\right] - \log p(D) \tag{4}$$

Since $\log p(D)$ is a constant and cannot be directly optimized, the evidence lower bound (ELBO) is typically maximized instead. The ELBO is defined as:

$$\mathcal{L}_{\mathrm{EB}} = E_{q(\theta)}[\log p(D|\, \theta)] - KL(q(\theta)\|p(\theta)) \tag{5}$$

Here, the first term, the expected likelihood, measures how well the data is fit under the variational distribution; The second term, the KL divergence, measures the difference between the variational distribution n and the prior distribution. By maximizing the ELBO, the variational distribution $q(\theta)$ can be effectively optimized, leading to an approximate posterior distribution of the model parameters.

3. Proposed method

The overall framework of MSAUQ, as illustrated in figure 1, comprises three main components: sample construction, model training, and UQ, with a total of 80,225 parameters. The internal architecture of the model integrates an encoder, built upon multi-scale attention modules, and a predictor, composed of linear layers. The encoder is designed to extract degradation-related features from the input samples, while the predictor maps these features to the RUL. During the prediction phase, Monte Carlo (MC) dropout is employed to perform multiple stochastic samplings of the model's predictions. Subsequently, KDE is utilized to estimate the PDF of the prediction outcomes.

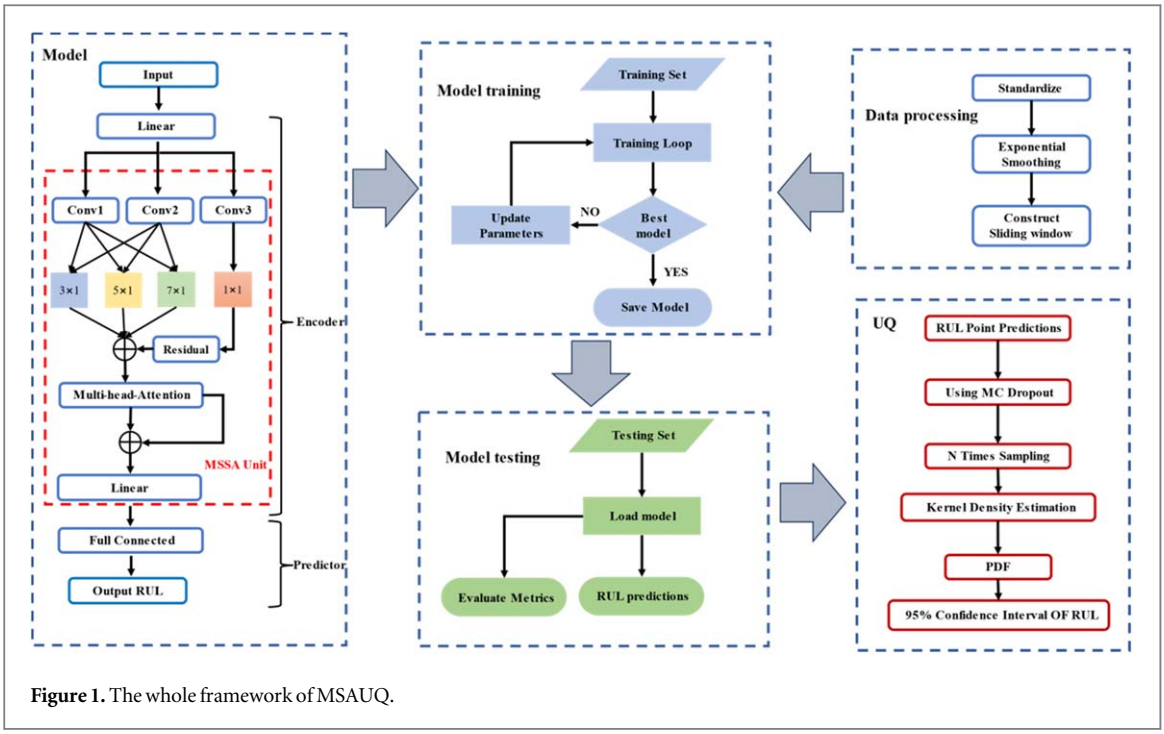


Figure 1. The whole framework of MSAUQ.

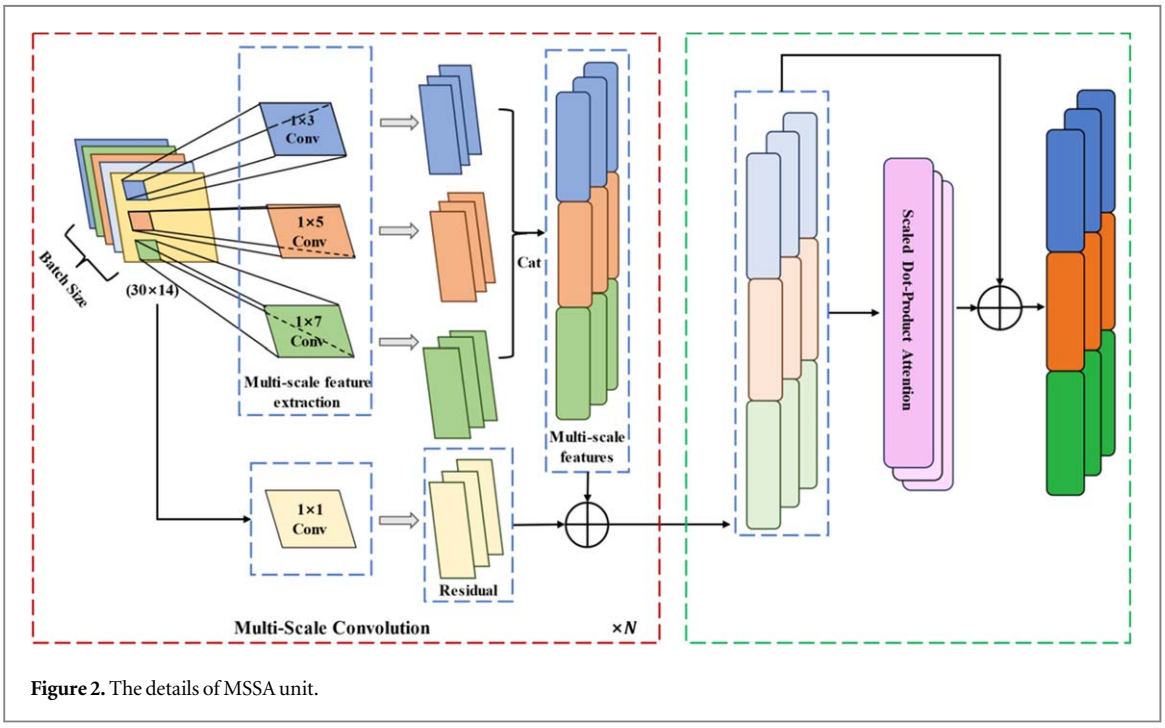


Figure 2. The details of MSSA unit.

3.1. MSSA unit

The overall structure of the proposed MSSA Unit is illustrated in figure 2. It consists of N multi-scale convolutional modules and a self-attention layer. Given an input sample x, it is first processed by the multi-scale convolutional modules to capture features at different temporal scales. Each multi-scale convolutional module performs multi-scale convolution operations, including convolution kernels of various sizes ($k \in \{3, 5, 7\}$).

For the input x_n to the nth multi-scale convolutional module, multiple convolution kernels of size k are applied to x_n, generating feature representations at different scales. Each convolution operation undergoes Batch normalization (BatchNorm), ReLU activation, and Dropout, which can be expressed as:

$$\mathrm{Conv}_k(x_n) = \mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{Conv1d}(x_n, k))) \tag{6}$$

The outputs of these convolutions are then concatenated along the channel dimension:

$$\text{Multi_scale_out} = \text{concat}(\text{Conv}_3(x_n), \text{Conv}_5(x_n), \text{Conv}_7(x_n)) \tag{7}$$

Next, residual connections are applied to the input x_n, where the original input x_n is added to the output of the multi-scale convolution. When the input feature map of the module does not match the concatenated multi-scale features in the channel dimension, we apply a linear projection on the residual branch to align the dimensionality. This projection is implemented using a 1×1 convolution, which effectively adjusts the number of channels without altering the spatial resolution (i.e., the sequence length) of the feature map. If the input and output channel dimensions are already consistent, the projection is replaced by an identity mapping. This design ensures dimensional consistency within the residual structure, thereby enabling effective feature fusion and stable model learning.

For example, if the concatenated multi-scale feature tensor has a shape of [8, 30, 96], a 1×1 convolution can be used to reduce the channel dimension to match a target shape of [8, 30, 64]. The 1×1 convolution operation in this case can be mathematically represented as:

$$\text{residual} = \text{Conv1d}(x_n, 1) \tag{8}$$

The final output of this process is:

$$x_{n+1} = \text{ReLU}(\text{Multi_scale_out} + \text{residual}) \tag{9}$$

The output x_n of the Nth layer is the feature map obtained after N iterations of multi-scale convolution and residual connections.

Next, positional encoding is added to x_N. Positional encoding is generated using sine and cosine functions to encode each timestep:

$$\text{PE}[i, 2j] = \sin\left(\frac{i}{10000^{\frac{2j}{\text{embed_dim}}}}\right), \quad \text{PE}[i, 2j+1] = \cos\left(\frac{i}{10000^{\frac{2j}{\text{embed_dim}}}}\right) \tag{10}$$

Where embed_dim is a predefined embedding dimension. The positional encoding is then added to the embedded features:

$$x_{pos} = x_{n+1} + \text{PE} \tag{11}$$

This ensures that each timestep's features contain both the information and its position in the sequence. The positional encoding-enhanced features x_{pos} are then fed into the self-attention mechanism. In the self-attention mechanism, the Query (Q), Key (K), and Value (V) are obtained through linear transformations of the input x_{pos}:

$$Q = W_Q x_{pos}, \quad K = W_K x_{pos}, \quad V = W_V x_{pos} \tag{12}$$

Attention weights are computed, and the values are weighted and summed:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{13}$$

where d_k is the dimensionality of the Key and Query. The output of the attention mechanism undergoes layer normalization and Dropout:

$$x_{attn} = \text{LayerNorm}(x_{pos} + \text{Dropout}(\text{Attention}(Q, K, V))) \tag{14}$$

Finally, a residual connection is used to combine x_{attn} and x_{n+1}, producing the final output of the multi-scale self-attention module x_{MSattn}:

$$x_{MSattn} = x_{attn} + x_{n+1} \tag{15}$$

3.2. Predictor

For the output x_{MSattn} from MSAUQ, the predictor first applies a pooling operation:

$$x_{pool} = \text{GlobalAvgPool}(x_{MSattn}) \tag{16}$$

The pooled output x_{pool} is then passed through two fully connected layers for regression, yielding the predicted RUL value $\hat{y}$:

$$\hat{y} = \text{Linear}(\text{ReLU}(\text{Linear}(x_{pool}))) \tag{17}$$

3.3. UQ under the approximate bayesian framework

In deep learning models, it is often infeasible to directly compute the analytical form of the posterior distribution $p(\theta|\ D)$. To address this, MC Dropout [38] is employed as an approximate Bayesian method, introducing uncertainty into the model weights by randomly masking a subset of neurons. This allows for variational distribution $q(\theta)$ to be approximately sampled, thereby enabling the modeling and quantification of model uncertainty. Combined with KDE, the Monte Carlo sampling results can be further used to estimate a continuous distribution, ultimately obtaining the PDF for RUL predictions.

First, by applying a random masking operation to the weight matrix θ, a randomized weight matrix is generated:

$$\tilde{\theta} = \theta \odot z,\ \ z \sim \text{Bernoulli}(p) \tag{18}$$

Where $\odot$ denotes element-wise multiplication, and z is a random variable sampled from a Bernoulli distribution, representing the activation state of neurons. p is the probability of retaining a neuron.

During each forward pass, a sample of network weights $\tilde{\theta}_i$ can be regarded as being drawn from the variational distribution $q(\theta)$. Assuming G samples are drawn for the input features x, the corresponding predictions are:

$$\hat{y}_i = f(x;\ \tilde{\theta}_i),\ \ \tilde{\theta}_i \sim q(\theta) \tag{19}$$

Through this process, G independent prediction samples $\{\hat{y}_i\}_{i=1}^{G}$ are generated, reflecting the sources of uncertainty in RUL prediction.

By performing statistical analysis on the sampling results $\{\hat{y}_i\}_{i=1}^{G}$, the mean and uncertainty of the RUL predictions can be quantified. To model the discrete sampling results $\{\hat{y}_i\}_{i=1}^{G}$ as a continuous distribution, KDE is introduced to construct the PDF of RUL predictions.

Using a kernel function $K(\cdot)$, the sampling results are smoothed, and the PDF is defined as:

$$\hat{p}(y) = \frac{1}{Gh}\sum_{i=1}^{N} K\left(\frac{y - \hat{y}_i}{h}\right) \tag{20}$$

where h is the bandwidth parameter controlling the degree of smoothing, and $K(\cdot)$ is the kernel function, typically a Gaussian kernel defined as:

$$K(u) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right) \tag{21}$$

Substituting the Gaussian kernel into the KDE formula, the result becomes:

$$\hat{p}(y) = \frac{1}{Gh\sqrt{2\pi}}\sum_{i=1}^{N}\exp\left(-\frac{(y - \hat{y}_i)^2}{2h^2}\right) \tag{22}$$

In practice, due to the finite number of samples and the choice of kernel function, the integral of the PDF $\hat{p}(y)$ generated by KDE may not strictly equal 1. Therefore, normalization is required to ensure that $\hat{p}(y)$ satisfies the definition of a PDF. First, the normalization factor is calculated as:

$$Z = \int_{-\infty}^{\infty} \hat{p}(y)\ dy \tag{23}$$

Since KDE is in a discrete form, the integral can be approximated as:

$$Z \approx \sum_{j} \hat{p}(y_j)\Delta y \tag{24}$$

Where Δy is the step size of the integration.

The PDF $\hat{p}(y)$ is then normalized to ensure it integrates to 1, resulting in the final normalized PDF for the RUL prediction:

$$\hat{p}^*(y) = \frac{\hat{p}(y)}{Z} = \frac{\hat{p}(y)}{\sum_j \hat{p}(y_j)\Delta y} \tag{25}$$

4. Experiment and analyze

4.1. Datasets description

The C-MAPSS dataset [39] used in this experiment contains sensor data from multiple engines operating under different conditions and fault modes, providing a complete degradation process. The dataset is divided into four

Table 1. Information of C-MAPSS dataset.

C-MAPSS	FD001	FD002	FD003	FD004
Engine units in training set	100	260	100	249
Engine units in testing set	100	259	100	248
Operating conditions	1	6	1	6
Fault modes	1	1	2	2

subsets: FD001, FD002, FD003, and FD004. Each subset includes a training set and a testing set. The training set contains measurements from 21 sensors that are most affected by engine faults. These datasets cover single or multiple operating conditions and fault modes, with each engine's data stored as a time series until a fault occurs. Table 1 summarizes the relevant information about the C-MAPSS dataset.

4.2. Data pre-processing

To improve data processing efficiency and conserve computational resources, this study selects 14 key sensors from the 21 sensors that significantly impact engine degradation as the primary focus. The remaining 7 sensors exhibit minimal variation during the degradation process and contribute less to the prediction results; therefore, they are excluded from the analysis. The final selected sensor numbers are: 2, 3, 4, 7, 8, 9, 11, 12, 13, 14, 15, 17, 20, and 21 [40]. The data from these sensors are used as inputs to analyze and predict the main factors influencing engine degradation. During the early stages of degradation, the degradation curve is relatively gradual and lacks a clear trend. Therefore, the early degradation RUL value is set to 125.

The RUL in the training set is calculated as the difference between the current operational cycle and the maximum cycle in the equipment's life, where $Cycle_{\max 1}$ represents the maximum operational cycle in the training set, i.e., the final cycle at which the equipment stops functioning, and $Cycle_{now1}$ represents the current recorded operational cycle. The calculation process is shown below:

$$RUL_{train} = Cycle_{\max 1} - Cycle_{now1} \tag{26}$$

The RUL in the test set is calculated based on the adjustment relationship between the true RUL value and the current operational cycle, as shown below:

$$RUL_{test} = RUL + Cycle_{\max 2} - Cycle_{now2} \tag{27}$$

Here, RUL is the true remaining useful life value directly provided in the test set, $Cycle_{\max 2}$ represents the maximum operational cycle achievable by the equipment in the test set, and $Cycle_{now2}$ is the current operational cycle in the test set.

To ensure that the sensor data values are not affected by their magnitudes and to highlight the impact of different operating conditions on the data, this study performs data normalization by grouping operating conditions. The normalization process is based on Z-Score normalization, and the formulas for the training and test sets are as follows:

$$Z_{group} = \frac{x_{group} - u_{group}}{\sigma_{group}} \tag{28}$$

where Z_{group} and x_{group} represent the normalized and original data in both the training and test sets, respectively. u_{group} is the mean of the grouped data in the training set, and σ_{group} is the standard deviation of the grouped data in the training set.

In the C-MAPSS dataset, sensor measurements may be affected by environmental noise, leading to significant fluctuations in the raw data. To reduce noise interference, improve the signal-to-noise ratio, and better capture the degradation trends, this study applies exponential smoothing after grouped normalization to improve data stability and smoothness. The smoothing operation is represented as follows:

$$S_t = \alpha \cdot x_t + (1 - \alpha) \cdot S_{t-1} \tag{29}$$

Here, S_t is the smoothed value at time t, calculated as a weighted combination of the current raw value x_t and the smoothed value S_{t-1} from the previous time step. The parameter α controls the weight distribution: a larger α gives more weight to current data changes, while a smaller α emphasizes historical data, resulting in a smoother data curve.

In aircraft engine RUL prediction, time series data also plays a crucial role, as its temporal characteristics reflect the degradation patterns of the equipment. However, directly using the complete time series for training significantly increases computational costs and makes it challenging to capture short-term dependencies, thus

hindering the accurate extraction of degradation features. To address this issue, a fixed-length sliding window approach is adopted to divide the original time series into overlapping subsequences. By continuously sliding the window, the model receives a steady stream of input data. The sliding window length L is set to 30, and the stride S is set to 1. Starting from the beginning of the sequence, the window extracts L consecutive time steps of data to form a subsequence, then moves S time steps along the time axis to generate the next subsequence. This process is repeated until the window reaches the end of the time series. Using the sliding window, the original sequence is transformed into N groups of subsequences of equal length. Each group has a shape of (L, M), where M is the number of selected sensor features, set to 14.

4.3. Performance metrics

To quantify the performance of MSAUQ and compare the advantages of different models, this study employs two evaluation metrics as the assessment criteria:

1. Root Mean Square Error (RMSE)

   MSAUQ uses the difference between the predicted and true values as an evaluation metric. The RMSE is calculated as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{30}$$

   where n is the number of test samples, y_i is the true value of the ith sample, and $\hat{y}_i$ is the predicted value of the ith sample. A lower RMSE value indicates that the predicted values are closer to the true values, with smaller errors. Conversely, a higher RMSE value indicates a larger deviation between the predicted and true values, with greater errors. RMSE reflects the overall error of MSAUQ.

2. Score

The Score metric is more aligned with practical application scenarios, as it evaluates MSAUQ's ability to perceive risks when predicting the remaining useful life (RUL). A lower score indicates that MSAUQ is better at accurately predicting equipment degradation, especially in handling different types of deviations, such as over-prediction and under-prediction. The calculation of the Score is shown below:

$$Score = \sum_{i=1}^{n} \begin{cases} e^{-\left(\frac{\hat{y}_i - y_i}{13}\right)} - 1, & \hat{y}_i - y_i < 0 \\ e^{\left(\frac{\hat{y}_i - y_i}{10}\right)} - 1, & \hat{y}_i - y_i > 0 \end{cases} \tag{31}$$

where n is the number of test samples, y_i is the true value of the ith sample, and $\hat{y}_i$ is the predicted value of the ith sample. When $\hat{y}_i$ is less than y_i (underestimating RUL), it could lead to premature maintenance or replacement, and the penalty denominator is 13 with a negative exponential factor to emphasize the impact of underestimation. When $\hat{y}_i$ is greater than y_i (overestimating RUL), it could increase the risk of equipment failure. In this case, the penalty denominator is 10, with a relatively lower weight for overestimation impacts.

In figure 3, a comparison of RMSE and Score is presented.

4.4. Experiment result

4.4.1. Prediction results on the test set

During MSAUQ training process, the RMSE value is used as the evaluation metric. The RMSE value of the current training results is continuously calculated, and when the RMSE reaches its minimum, the corresponding model is saved as the best weight file. During prediction, the weight file and model are loaded to produce the prediction results for the four datasets, which are compared with the true values, as shown in figures 4–7.

Based on the RUL prediction results for the four datasets (FD001, FD002, FD003, FD004), MSAUQ demonstrates outstanding performance across all datasets, showcasing strong predictive capabilities. For the FD001 dataset, due to its single operating condition and single fault type, MSAUQ is able to precisely capture the trend of RUL changes. The predicted curve aligns closely with the true values, especially in the low and medium RUL ranges, where MSAUQ exhibits exceptional accuracy. For the FD002 dataset, despite its multiple operating conditions and single fault mode, MSAUQ consistently delivers stable predictions. Notably, in the low RUL region, the predicted values are almost perfectly aligned with the true values, demonstrating excellent adaptability to complex operating conditions. For the FD003 dataset, the single operating condition with multiple fault types does not affect MSAUQ's performance. MSAUQ accurately predicts RUL under different fault modes, and the predicted curve closely follows the trend of the true values. In the FD004 dataset, despite its

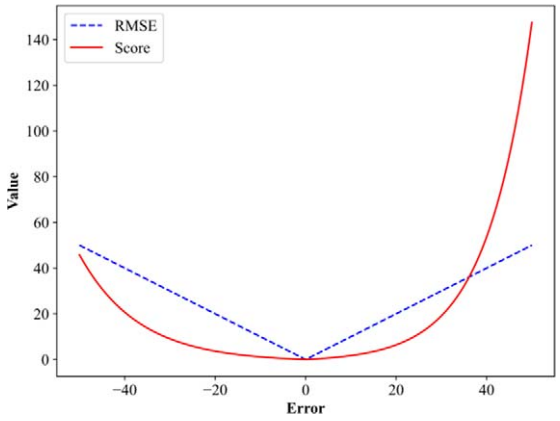


Figure 3. Comparison of the RMSE and score.

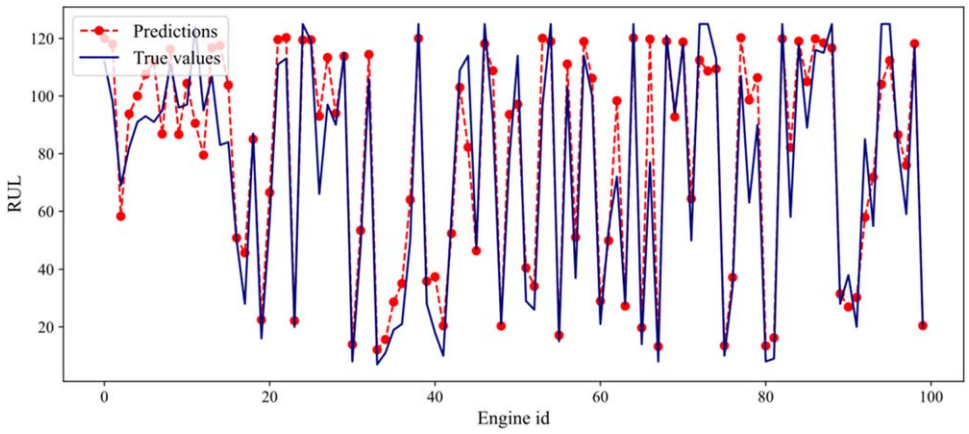


Figure 4. Comparison of predicted and true values for FD001.

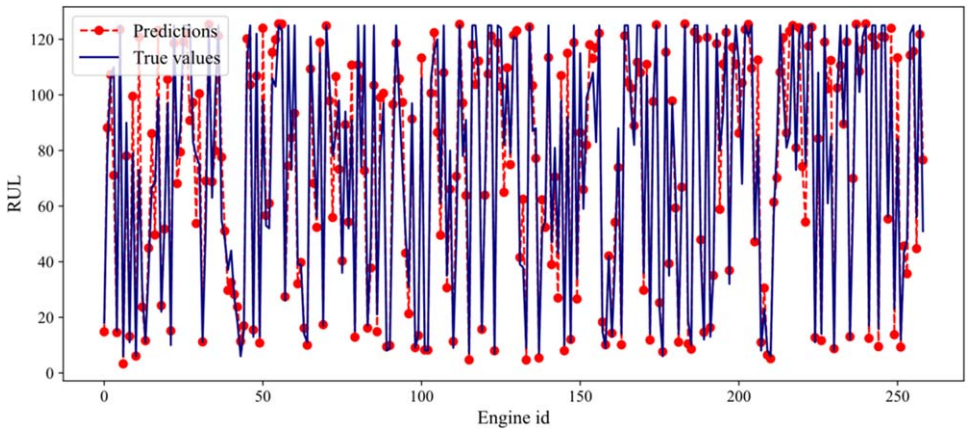


Figure 5. Comparison of predicted and true values for FD002.

high complexity with multiple operating conditions and multiple fault modes, MSAUQ remains stable in both low and medium RUL ranges. The prediction results closely reflect actual value.

In summary, MSAUQ achieves a high level of accuracy across all datasets. Additionally, during severe degradation, MSAUQ is able to provide even more precise RUL predictions.

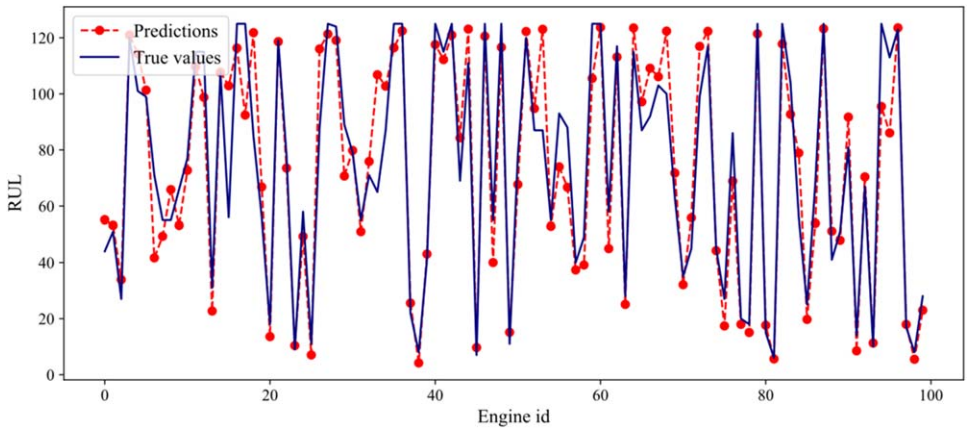


Figure 6. Comparison of predicted and true values for FD003.

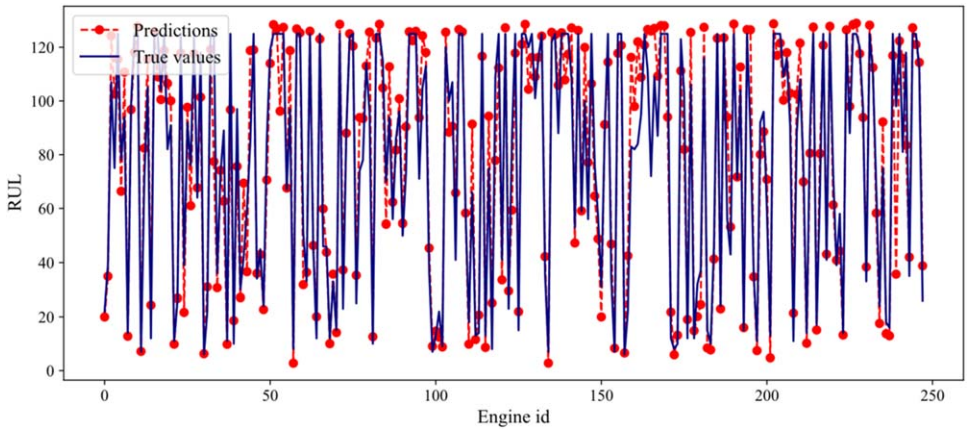


Figure 7. Comparison of predicted and true values for FD004.

4.4.2. UQ

With regard to the sampling number G discussed in section 3.3., this study finds that increasing G generally enhances the stability and accuracy of UQ. A small value of G may result in highly volatile uncertainty estimates, which fail to accurately reflect the model's true confidence level. On the other hand, the computational cost of UQ scales directly with the sampling number, as each additional stochastic forward pass increases the time required for online prediction or evaluation. Therefore, although a large G may yield more precise UQ, it could also render the model impractical for applications with strict constraints on computational resources or real-time responsiveness. Based on these considerations, we set $G = 100$ to strike a balance between computational efficiency and the stability of the UQ.

We selected one engine ID from each of the four datasets to predict its degradation curve. Additionally, the CIs for each prediction point is marked in the figure, as shown in figure 8.

As shown in the figure 8, the predicted RUL degradation trend generally aligns with the actual degradation curve. Furthermore, as the engine degrades, the width of the CIs decreases, indicating that MSAUQ's predictive capability and reliability improve in the later stages of degradation. Next, points corresponding to 80%, 60%, 40%, and 20% of RUL on curve in figure 8 are selected to display the PDF estimation results for further analysis, as shown below.

In figures 9–12 the four subplots for each dataset, from left to right, correspond to the PDF of points at 80%, 60%, 40%, and 20% of the remaining RUL, respectively.

As the RUL decreases, the peak of the PDF becomes higher while the bandwidth becomes smaller, since the area of the green shaded region remains constant at 1. This indicates that the confidence in the prediction results increases as the RUL decreases.

To further evaluate the overall performance of the proposed UQ framework in terms of predictive accuracy and computational efficiency, we conducted a detailed comparison with a baseline model based on Bayesian

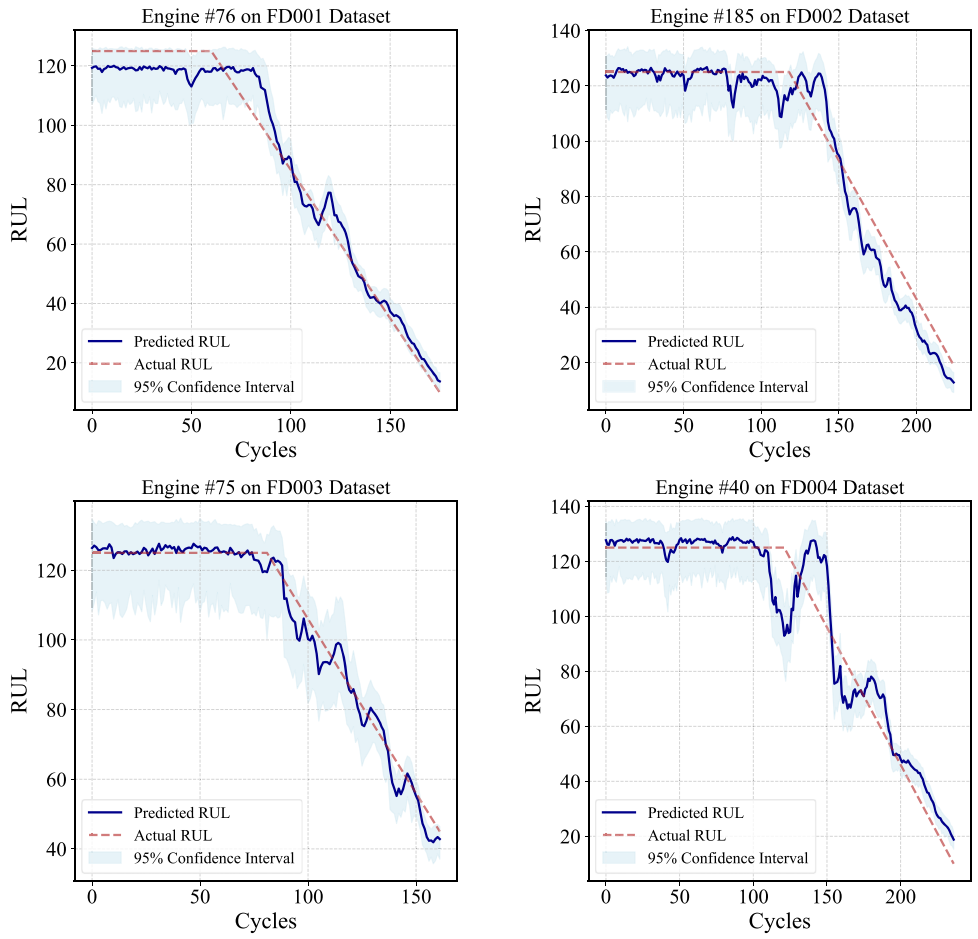


Figure 8. The RUL predictions of randomly selected engines from the four datasets with CIs.

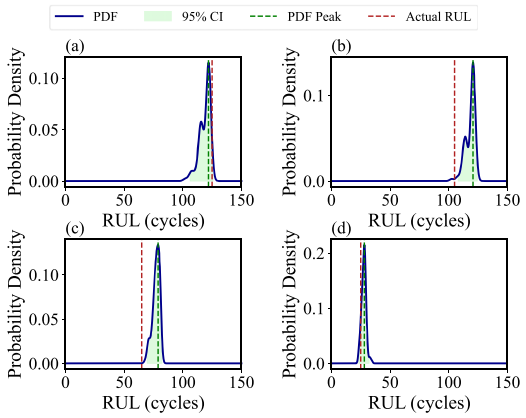


Figure 9. PDF for the four points on FD001.

Neural Networks (BNNs) on the FD004 subset of the C-MAPSS dataset. To ensure a fair comparison, both approaches were implemented under identical experimental settings: they shared the same backbone network architecture, were trained on the same FD004 dataset with identical data preprocessing procedures, and used the same optimizer (Adam) with 50 training epochs. The experimental results are presented in table 2.

As shown in the table, the proposed method outperforms the BNN baseline not only in terms of prediction accuracy but also in training and inference efficiency. These results demonstrate that our method is more lightweight and efficient than the BNN-based approach, achieving a better trade-off between UQ capability, prediction accuracy, and runtime performance.

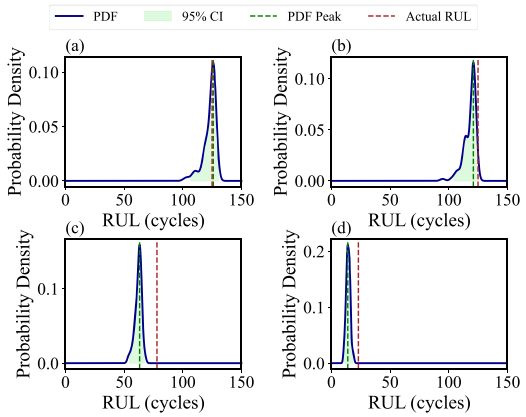


Figure 10. PDF for the four points on FD002.

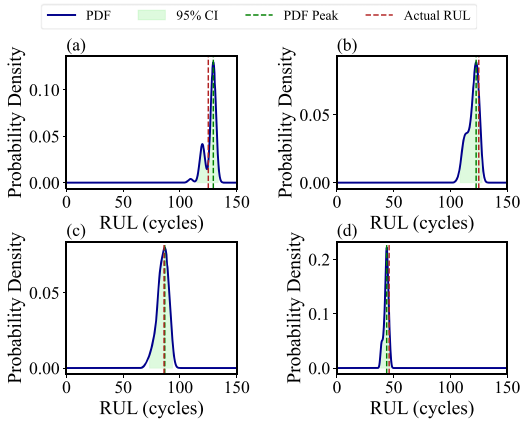


Figure 11. PDF for the four points on FD003.

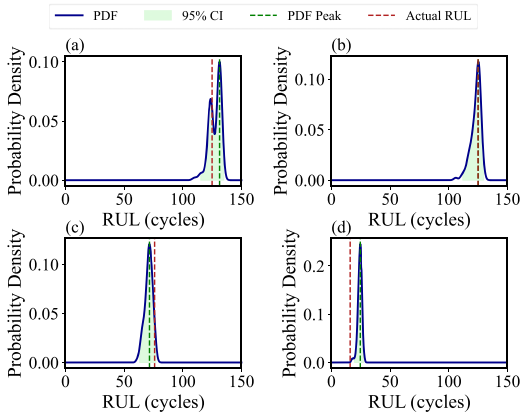


Figure 12. PDF for the four points on FD004.

4.5. Ablation experiment

The proposed model combines multi-scale convolution and multi-head attention mechanisms. To verify the contribution of each module, this paper conducted an ablation experiment. While keeping other variables unchanged, three simplified models were constructed based on the original model: Single-Scale Convolution, Multi-Scale Convolution, and Multi-Head Attention Mechanism. Following MSAUQ, multiple rounds of training and prediction were conducted, with adjustments to learning rates and other parameters. The best

Table 2. Performance comparison of proposed method and bayesian baseline model (FD004 dataset).

Metric	Proposed method	Bayesian baseline model
RMSE (FD004)	13.85	17.43
Training Time (50 epochs) (s)	119.17	147.36
Inference Time per Forward Pass (single sample) (ms)	1.905	3.657
UQ Inference Time (single sample, 100 passes) (ms)	190.5	365.7

Table 3. Comparation of four methods.

Method	C-MAPSS							
	FD001		FD002		FD003		FD004	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
Single-Scale Convolution	22.61	1853.25	20.11	58455.58	22.45	1185.76	22.83	3095.47
Muti-Scale Convolution	15.79	350.70	17.78	4707.39	15.14	347.39	21.90	8576.95
Muti-Head Attention Mechanism	13.92	316.75	14.87	1576.22	14.51	302.99	15.43	10453.04
Proposed Method (MSAUQ)	**13.74**	**255.89**	**14.19**	**1173.83**	**13.68**	**286.37**	**13.85**	**1222.37**

results for the three models across multiple training and prediction runs were compared with MSAUQ, as shown in table 3.

It can be observed that the performance of different modules on the C-MAPSS dataset varies significantly. Single-scale convolution, constrained by its limited receptive field, lacks the capability to model multi-condition degradation patterns effectively (e.g., FD002/FD004 Scores are 58455.58/3095.47, respectively). In contrast, multi-scale convolution improves local representation through multi-receptive field feature extraction, but its inability to model global temporal dependencies limits its performance on complex datasets. Multi-head self-attention excels at global temporal modeling (with FD001/FD003 performance approaching that of MSAUQ) but fails to handle multi-condition predictions due to insufficient local feature capture (FD004 Score = 10453.04).

In comparison, MSAUQ integrates the local feature extraction capability of multi-scale convolution with the global dependency modeling capability of multi-head self-attention, achieving coordinated optimization of spatiotemporal features. This results in the best RMSE and Score across all four subsets (e.g., FD002/FD004 Scores = 1173.83/1222.37), validating the effectiveness of multimodal feature fusion in modeling complex degradation patterns.

Meanwhile, to systematically evaluate the impact of different positional encoding (PE) strategies on the performance of RUL prediction, we conducted a series of comparative experiments on the FD004 subset of the C-MAPSS dataset. Positional encoding serves as a critical component in self-attention mechanisms for modeling temporal dependencies, which is particularly important in tasks like RUL prediction where strong sequential correlations exist. Therefore, this study compares the effectiveness of four distinct positional encoding methods under an identical network architecture.

The first method is No Positional Encoding (No PE), where the model relies entirely on its convolutional and attention mechanisms to implicitly capture temporal patterns, without incorporating any explicit positional cues. The second method is Learnable Absolute Positional Encoding (Learnable PE), which assigns each time step a unique, trainable position vector, allowing the model to learn optimal representations of positional context during training. The third method is Embedding-based Learnable Positional Encoding (Embedding PE), in which time steps are treated as indices into a trainable embedding layer that outputs discrete positional vectors. This approach offers greater flexibility but may be more susceptible to overfitting the training distribution. The fourth method, adopted in this study, is Fixed Sinusoidal Positional Encoding (Sinusoidal PE). This non-learnable strategy encodes positions using sinusoidal functions of varying frequencies, providing global, multi-scale temporal information without introducing additional parameters.

To ensure a fair comparison, all experiments were conducted under a unified neural network architecture, with the only difference being the positional encoding strategy. The network backbone consists of multi-scale convolutional modules and a self-attention mechanism, and all hyperparameters were kept consistent across experiments. Model performance was evaluated using RMSE as the primary metric, and the experimental results are presented in figure 13.

The experimental results clearly demonstrate that the method employing Sinusoidal PE achieves the best performance in terms of RMSE, outperforming the other three positional encoding strategies. This finding

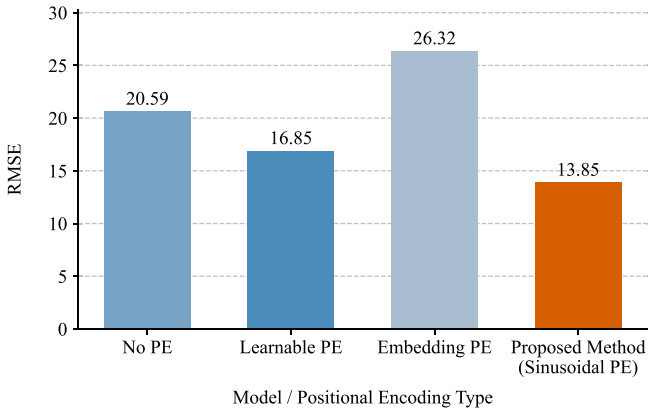


Figure 13. Comparison of different positional encoding strategies on the FD004 dataset.

indicates that explicitly modeling temporal dependencies is beneficial for improving model performance in RUL prediction tasks. Compared to learnable positional encodings, sinusoidal encoding offers stronger generalization capability due to its fixed and frequency-based structure, which does not rely on the distribution of the training data. Moreover, the design of multiple frequency components enables the model to effectively capture degradation trends and temporal rhythms across different time scales, thereby enhancing predictive accuracy. In summary, the Sinusoidal PE adopted in this study exhibits superior temporal modeling ability and robust generalization performance in the context of RUL prediction for rotating machinery.

4.6. Hyperparameter sensitivity

To understand MSAUQ's dependence on key hyperparameters and its performance stability when these hyperparameters are changed, this paper conducted extensive experiments to perform a detailed analysis of the sensitivity to convolutional scale combinations, convolutional depth and smoothing factor.

4.6.1. Sensitivity to different scales

By altering different combinations of convolutional scales to provide varying receptive fields, MSAUQ acquires different local features. The RMSE and Score values for different convolutional scale combinations are compared to test MSAUQ's sensitivity to these scales and to identify the optimal receptive field combination. The convolutional scale combinations used in this experiment are [1, 3, 5], [3, 5, 7], [5, 7, 9], and [7, 9, 11].

Based on figures 14 and 15, which show the RUL prediction performance of different convolutional scales ([1, 3, 5], [3, 5, 7], [5, 7, 9], and [7, 9, 11]) across the four datasets, it can be observed that the RMSE values for the four scale combinations across the datasets show minimal differences. Similarly, the penalty scores for the smaller datasets, FD001 and FD003, exhibit almost no significant differences. However, for the larger datasets, FD002 and FD004, the penalty scores show noticeable differences due to the exponential operations involved, as reflected in the figures. Nevertheless, these differences still remain within the same order of magnitude.

Therefore, it can be concluded that the choice of convolutional scale has a limited impact on MSAUQ's performance. Within the allowable range, the combination [3, 5, 7] demonstrates the most stable performance among the four combinations, especially for the most complex dataset, FD004, where it delivers the best performance. This is the reason why the MSAUQ model proposed in this study adopts the convolutional scale combination of [3, 5, 7].

4.6.2. Sensitivity to the number of convolutions

Increasing the depth of convolution can enhance MSAUQ's ability to extract higher-order and more complex features. Adding more convolutional layers can provide stronger representation capabilities. However, MSAUQ's performance does not necessarily improve with an increasing number of convolutional layers. A higher number of convolution operations also leads to increased computational cost and longer processing time. In this study, while keeping other variables constant, the number of convolutional layers N is set to 1, 2, 3, and 4, respectively, followed by training and evaluation of the results. By comparing the RMSE and Score values under different convolutional depths, the sensitivity to the number of convolutions is analyzed.

As shown in figures 16 and 17, the RMSE and Score of different convolutional depths N on various datasets are presented. It can be observed that the number of convolutions has little impact on the RMSE values across datasets, with RMSE fluctuating within a small range. Similarly, the effect of convolutional depth on the Score for simpler datasets FD001 and FD003 is minimal, especially for FD001, where almost no changes are observed.

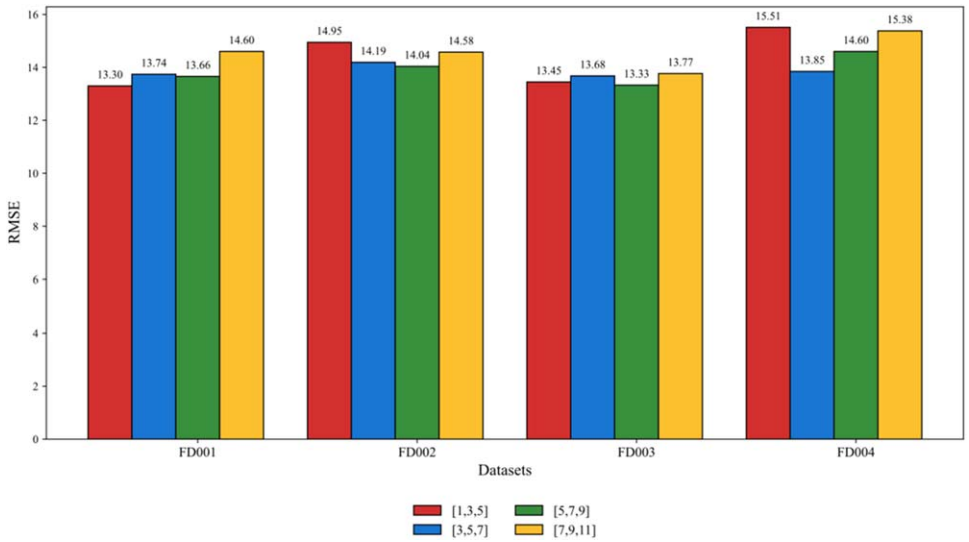


Figure 14. RMSE comparison across different scales.

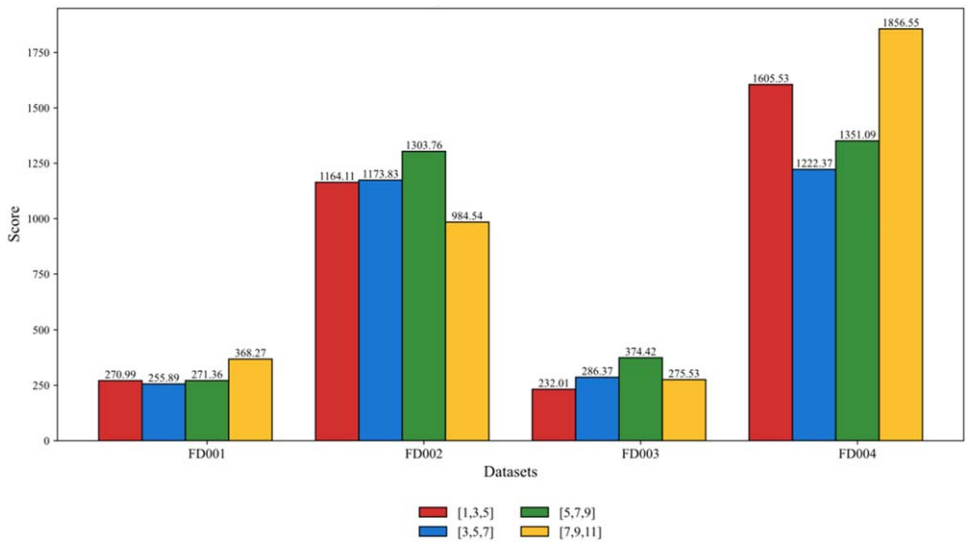


Figure 15. Score comparison across different scales.

However, as the complexity of the datasets increases, convolutional depth has a greater impact on the FD002 and FD004 datasets. When $N = 4$, both the RMSE and Score achieve optimal performance. When $N = 1$ or $N = 3$, the Score on FD004 is relatively high. When $N = 2$, although the overall RMSE is higher than the other three configurations and the Score for FD002 reaches its highest value, its Score on FD004 is close to the optimal level, with only a slight difference from the best value for FD002.

Compared to $N = 4$, while some performance improvement is observed, the computational cost and time overhead increase significantly. Therefore, it can be concluded that within the allowable range, MSAUQ's sensitivity to the number of convolutions is not very significant. In the MSAUQ model of this study, N is set to 2, which achieves good results across all four datasets while keeping computational and time costs within a reasonable range.

4.6.3. Sensitivity to the smoothing factor

To evaluate the performance and robustness of the proposed model under different noise levels, experiments were conducted on the FD004 subset of the C-MAPSS dataset. Specifically, we applied exponential smoothing to the input data and varied the smoothing factor α from 0.1 to 1.0 to simulate different levels of signal smoothness. The setting of $\alpha = 1.0$ corresponds to the use of raw sensor data without any smoothing, allowing us to directly assess the model's baseline performance in noisy environments.

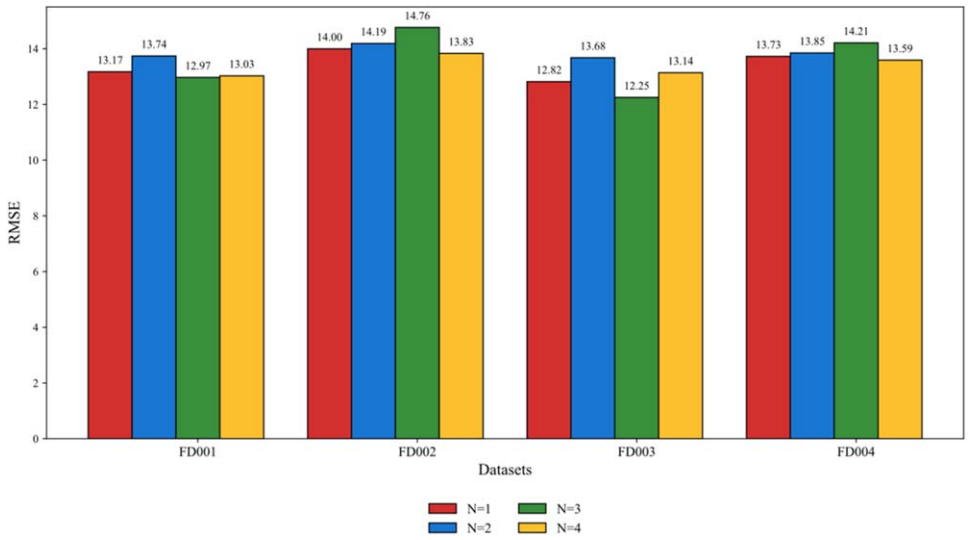


Figure 16. RMSE comparison across different number of convolutions.

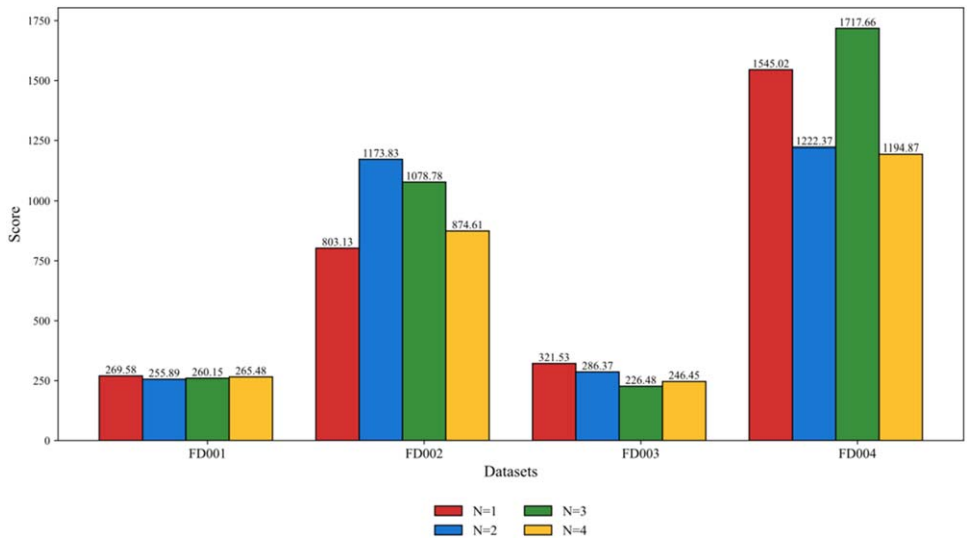


Figure 17. Score comparison across different number of convolutions.

As shown in figure 18, the experimental results indicate that when directly processing raw noisy data ($\alpha = 1.0$), the model achieves a RMSE of 16.18. This demonstrates MSAUQ's inherent robustness and its ability to handle noise commonly encountered in real-world industrial scenarios. Further experiments reveal that incorporating moderate smoothing can enhance model performance: when the smoothing factor is set to $\alpha = 0.2$, the RMSE is reduced to a best value of 13.85. These findings suggest that while the model is resilient to noise, appropriate data preprocessing—such as moderate smoothing to suppress noise—can facilitate more effective extraction of key degradation features and thereby improve predictive accuracy.

From the above sensitivity analysis of convolutional scale combinations, convolutional depth and smoothing factor, it can be concluded that when the convolutional scale is set to [3, 5, 7] and N is set to 2, and α is set to 0.2, MSAUQ achieves the optimal hyperparameters within the allowable range. Based on this configuration, the experiments in this paper proceed with the visualization of prediction results and the quantification of uncertainty in point predictions.

4.7. Comparison with previous work

We conducted a detailed comparison between the proposed method and recently published methods on the C-MAPSS dataset, with the results shown in table 4. By comparing the RMSE and Score metrics across the

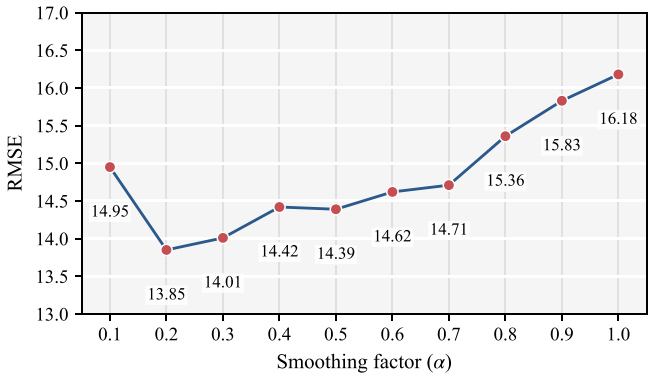


Figure 18. Effect of exponential smoothing factor (α) on RMSE (FD004).

Table 4. Comparison with existing studies.

Method	Year	FD001		FD002		FD003		FD004	
		RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
MSIDSN [41]	2023	11.74	205.55	18.26	2046.65	12.04	196.42	22.48	2910.73
WD-MSADA [42]	2023	15.09	485.00	16.38	2156.00	14.86	413.00	19.49	2594.00
STRUL [43]	2023	12.85	224.00	19.24	1950.00	13.74	252.00	22.34	3080.00
AttnPINN [44]	2023	16.89	523.00	16.32	1479.00	17.75	1194.00	18.37	2059.00
CNN-Wiener [45]	2023	11.68	196.00	21.22	2200.00	12.55	228.00	24.43	3800.00
DVGTformer [46]	2023	**11.33**	**179.75**	14.28	**797.26**	11.89	254.55	15.50	**1107.5**
FedCov [47]	2024	25.97	293.33	30.35	1238.73	25.94	318.85	30.97	1321.76
MSTSDN [2]	2024	13.67	246.01	16.28	1342.59	13.66	258.92	17.33	1641.51
ARR [48]	2024	11.36	192.22	18.97	2433.15	**11.28**	**133.41**	20.69	2842.44
MSAUQ(ours)	2025	13.74	255.89	**14.19**	1173.83	13.68	286.37	**13.85**	1222.37

FD001 to FD004 subsets, it is evident that the proposed method demonstrates optimal performance on several subsets. For instance, on the FD002 and FD004 subsets, our method achieved the lowest RMSE 14.19 and 13.85.

5. Conclusions

This paper proposes the MSAUQ method, which combines multi-scale learning and self-attention mechanisms to achieve fusion of multi-scale features. It effectively addresses the limitations of traditional RUL prediction methods in multi-scale feature extraction. Furthermore, through the use of an approximate Bayesian method and KDE, the method enables probabilistic RUL estimation and completes the UQ of prediction results.

Extensive experiments conducted on the C-MAPSS dataset demonstrate the performance advantages of the proposed method across all subsets. Specifically, under complex multi-condition and multi-fault modes (FD002 and FD004), the method achieved RMSE and Score values of 14.19 (1173.83) and 13.85 (1222.37), respectively. Additionally, ablation experiments and hyperparameter sensitivity analysis further validate the rationality and robustness of MSAUQ architecture, proving the contribution of multi-scale feature extraction and self-attention modules to enhancing prediction reliability.

Future work will focus on model lightweighting, the construction of cross-domain transfer learning frameworks [49–51], and the deep integration of multi-source heterogeneous data to enhance MSAUQ's generalization ability, computational efficiency, and adaptability to complex conditions. These efforts aim to facilitate the application of the proposed method to a broader range of industrial scenarios.

Acknowledgments

We sincerely appreciate the support from the National College Students' Innovation and Entrepreneurship Training Program of Inner Mongolia University. The project, 'Research on Semi - supervised Fuzzy Clustering Algorithm Based on Zonal - level Constraints' (Project No. 202410126022), has been fundamental to this research. Our heartfelt gratitude goes to the project supervisor, Bai Lan. Her profound knowledge and rich

experience have offered invaluable guidance throughout the research, leading us to clarify the research direction. Also, we are thankful to the project host, Fu Jinze, for his effective coordination, which has ensured the smooth progress of this study. In summary, this research is the fruit of our project team's joint efforts. We are truly grateful for the contributions of all involved.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

References

[1] Wang Y, Liu S, Lv S and Liu G 2024 Bearing remaining life prediction method based on ARAD -ELN and multi-stage wiener process *Meas. Sci. Technol.* **36** 016107
[2] Liu Z, Zheng X, Xue A and Ge M 2024 Multi-scale temporal-spatial feature-based hybrid deep neural network for remaining useful life prediction of aero-engine *ACS Omega* **9** 47410–27
[3] Zhou J, Qin Y, Chen D, Liu F and Qian Q 2022 Remaining useful life prediction of bearings by a new reinforced memory GRU network *Adv. Eng. Inf.* **53** 101682
[4] Keshun Y, Guangqi Q and Yingkui G 2024 A 3D attention-enhanced hybrid neural network for turbofan engine remaining life prediction using CNN and BiLSTM models *IEEE Sens. J.* **24** 21893–905
[5] Wu B, Zhang X, Shi H and Zeng J 2024 Failure mode division and remaining useful life prognostics of multi-indicator systems with multi-fault *Reliab. Eng. Syst. Saf.* **244** 109961
[6] Yin C, Li Y, Wang Y and Dong Y 2025 Physics-guided degradation trajectory modeling for remaining useful life prediction of rolling bearings *Mech. Syst. Signal. Pr.* **224** 112192
[7] Cheng Y, Hu K, Wu J, Zhu H and Lee C K M 2021 A deep learning-based two-stage prognostic approach for remaining useful life of rolling bearing *Applied Intelligence* **52** 1–16
[8] Fernández J, Corbetta M, Kulkarni C S, Chiachío J and Chiachío M 2024 Training of physics-informed Bayesian neural networks with ABC-SS for prognostic of Li-ion batteries *Comput. Ind.* **155** 104058
[9] Liu J 2021 Maintenance model of aircraft structure based on three-stage degradation process *Comput. Ind. Eng.* **157** 107335
[10] Wang Y, Li M, Zheng L, Shi M, Zheng Z and Pei X 2024 Phyformer: a degradation physics-informed self-data driven approach to machinery prognostics *Adv. Eng. Inf.* **62** 102772
[11] Li H, Zhang Z, Li T and Si X 2024 A review on physics-informed data-driven remaining useful life prediction: challenges and opportunities *Mech. Syst. Sig. Process.* **209** 111120
[12] Xiang S, Li P, Luo J and Qin Y 2025 Micro transfer learning mechanism for cross-domain equipment RUL prediction *IEEE Trans. Autom. Sci. Eng.* **22** 1460–70
[13] Liu S and Fan L 2022 An adaptive prediction approach for rolling bearing remaining useful life based on multistage model with three-source variability *Reliab. Eng. Syst. Saf.* **218** 108182
[14] Lv S, Liu S, Li H, Wang Y, Liu G and Dai W 2024 A hybrid method combining Lévy process and neural network for predicting remaining useful life of rotating machinery *Adv. Eng. Inf.* **61** 102490
[15] Wang Z, Ta Y, Cai W and Li Y 2023 Research on a remaining useful life prediction method for degradation angle identification two-stage degradation process *Mech. Syst. Sig. Process.* **184** 109747
[16] Wang H, Ma X and Zhao Y 2019 An improved Wiener process model with adaptive drift and diffusion for online remaining useful life prediction *Mech. Syst. Sig. Process.* **127** 370–87
[17] Zhang Y, Zhang C, Wang S, Dui H and Chen R 2024 Health indicators for remaining useful life prediction of complex systems based on long short-term memory network and improved particle filter *Reliab. Eng. Syst. Saf.* **241** 109666
[18] Wang Y, Chen L, Liu Y and Gao L 2021 Wavelet-prototypical network based on fusion of time and frequency domain for fault diagnosis *Sensors (Basel)* **21** 1483
[19] Ma P, Li G, Zhang H, Wang C and Li X 2024 Prediction of remaining useful life of rolling bearings based on multiscale efficient channel attention CNN and bidirectional GRU *IEEE Trans. Instrum. Meas.* **73** 1–13
[20] Chen X 2024 A novel transformer-based DL model enhanced by position-sensitive attention and gated hierarchical LSTM for aero-engine RUL prediction *Sci. Rep.* **14** 10061
[21] Jang J and Kim C O 2022 Siamese network-based health representation learning and robust reference-based remaining useful life prediction *IEEE Trans. Ind. Inf.* **18** 5264–74
[22] Zhang J *et al* 2024 A data-model interactive remaining useful life prediction approach of lithium-ion batteries based on PF-BiGRU-TSAM *IEEE Trans. Ind. Inf.* **20** 1144–54
[23] Shi J, Zhong J, Zhang Y, Xiao B, Xiao L and Zheng Y 2024 A dual attention LSTM lightweight model based on exponential smoothing for remaining useful life prediction *Reliab. Eng. Syst. Saf.* **243** 109821
[24] Zhang C, Zeng M, Fan J and Li X 2024 Application of residual structure time convolutional network based on attention mechanism in remaining useful life interval prediction of bearings *Sensors (Basel)* **24** 4132
[25] Costa N and Sánchez L 2022 Variational encoding approach for interpretable assessment of remaining useful life estimation *Reliab. Eng. Syst. Saf.* **222** 108353
[26] Wang Y and Liu S 2024 A multi scale meta-learning network for cross domain fault diagnosis with limited samples *J. Intell. Manuf.* **36** 2841–61
[27] Yao Y, Zhang S, Yang S and Gui G 2020 Learning attention representation with a multi-scale CNN for gear fault diagnosis under different working conditions *Sensors (Basel)* **20** 1233
[28] Chen X, Zhang B and Gao D 2020 Bearing fault diagnosis base on multi-scale CNN and LSTM model *J. Intell. Manuf.* **32** 971–87
[29] Niazi S G, Huang T, Zhou H, Bai S and Huang H-Z 2024 Multi-scale time series analysis using TT-ConvLSTM technique for bearing remaining useful life prediction *Mech. Syst. Signal. Pr.* **206** 110888
[30] Gao Z, Lei Y, Wu J, Jiang W, Wang Y and Ye X 2025 VSC-Net: versatile spatiotemporal convolution network with multi-sensor signals for remaining useful life prediction of mechanical systems *Information Fusion* **123** 103288

[31] Li Y, Du X, Wang X and Si S 2022 Industrial gearbox fault diagnosis based on multi-scale convolutional neural networks and thermal imaging *ISA Trans.* **129** 309–20
[32] Yang J, Peng Y, Xie J and Wang P 2022 Remaining useful life prediction method for bearings based on LSTM with uncertainty quantification *Sensors (Basel)* **22** 4549
[33] Ding P, Jia M, Ding Y, Cao Y, Zhuang J and Zhao X 2024 Machinery probabilistic few-shot prognostics considering prediction uncertainty *IEEE/ASME Trans. Mechatron.* **29** 106–18
[34] Wang H, Bai X and Tan J 2020 Uncertainty quantification of bearing remaining useful life based on convolutional neural network *IEEE Symp. Series on Computational Intelligence (ssci)* (IEEE) 2893–99
[35] Cao L, Zhang H, Meng Z and Wang X 2023 A parallel GRU with dual-stage attention mechanism model integrating uncertainty quantification for probabilistic RUL prediction of wind turbine bearings *Reliab. Eng. Syst. Saf.* **235** 109197
[36] Basora L, Viens A, Chao M A and Olive X 2023 A benchmark on uncertainty quantification for deep learning prognostics *Reliab. Eng. Syst. Saf.* **253** 108586
[37] Wang Y, Liu S, Lv S and Liu G 2025 Few-shot probabilistic RUL prediction with uncertainty quantification of slurry pumps *IEEE Sens. J.* **25** 6122–32
[38] Gal Y and Ghahramani Z 2016 Dropout as a bayesian approximation: representing model uncertainty in deep learning *International Conference on Machine Learning* 1050–9 (PMLR)
[39] Saxena A, Goebel K, Simon D and Eklund N 2008 Damage propagation modeling for aircraft engine run-to-failure simulation *International Conference on Prognostics and Health Management* 1–9 (IEEE)
[40] Zhang C, Lim P, Qin A K and Tan K C 2017 Multiobjective deep belief networks ensemble for remaining useful life estimation in prognostics *2017 IEEE Int. Conf. on Prognostics and Health Management (ICPHM)* 1–8 (https://doi.org/10.1109/ICPHM.2017.7998310)
[41] Zhao K, Jia Z, Jia F and Shao H 2023 Multi-scale integrated deep self-attention network for predicting remaining useful life of aero-engine *Eng. Appl. Artif. Intell.* **120** 105860
[42] Shi H, Huang C, Zhang X, Zhao J and Li S 2023 Wasserstein distance based multi-scale adversarial domain adaptation method for remaining useful life prediction *Applied Intelligence* **53** 3622–37
[43] Wang T, Li X, Wang W, Du J and Yang X J M 2023 A spatiotemporal feature learning-based RUL estimation method for predictive maintenance **214** 112824
[44] Liao X, Chen S, Wen P and Zhao S 2023 Remaining useful life with self-attention assisted physics-informed neural network *Adv. Eng. Inf.* **58** 102195
[45] Wen L, Su S, Wang B, Ge J, Gao L and Lin K 2023 A new multi-sensor fusion with hybrid convolutional neural network with wiener model for remaining useful life estimation *Eng. Appl. Artif. Intell.* **126** 106934
[46] Cai C, Fang Y, Liu W, Jin R, Cheng J and Chen Z 2024 FedCov: enhanced trustworthy federated learning for machine RUL prediction with continuous-to-discrete conversion *IEEE Trans. Ind. Inf.* **20** 12565–74
[47] Wang L, Cao H, Ye Z, Xu H and Yan J 2024 DVGTformer: a dual-view graph transformer to fuse multi-sensor signals for remaining useful life prediction *Mech. Syst. Signal. Pr.* **207** 110935
[48] Kim G, Choi J G and Lim S J 2024 Using transformer and a reweighting technique to develop a remaining useful life estimation method for turbofan engines *Eng. Appl. Artif. Intell.* **133** 108475
[49] Qian Q, Wen Q, Tang R and Qin Y 2025 DG-Softmax: a new domain generalization intelligent fault diagnosis method for planetary gearboxes *Reliab. Eng. Syst. Saf.* **260** 111057
[50] Qian Q, Pu H, Tu T and Qin Y 2024 Variance discrepancy representation: a vibration characteristic-guided distribution alignment metric for fault transfer diagnosis *Mech. Syst. Signal. Pr.* **217** 111544
[51] Qian Q, Luo J and Qin Y 2024 Heterogeneous federated domain generalization network with common representation learning for cross-load machinery fault diagnosis *IEEE Transactions on Systems, Man, and Cybernetics: Systems* **54** 5704–16